



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
23-05-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TAAS,TRAS,TEAS,TTAS,TOAS,TNAS,TMAS)

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **27** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

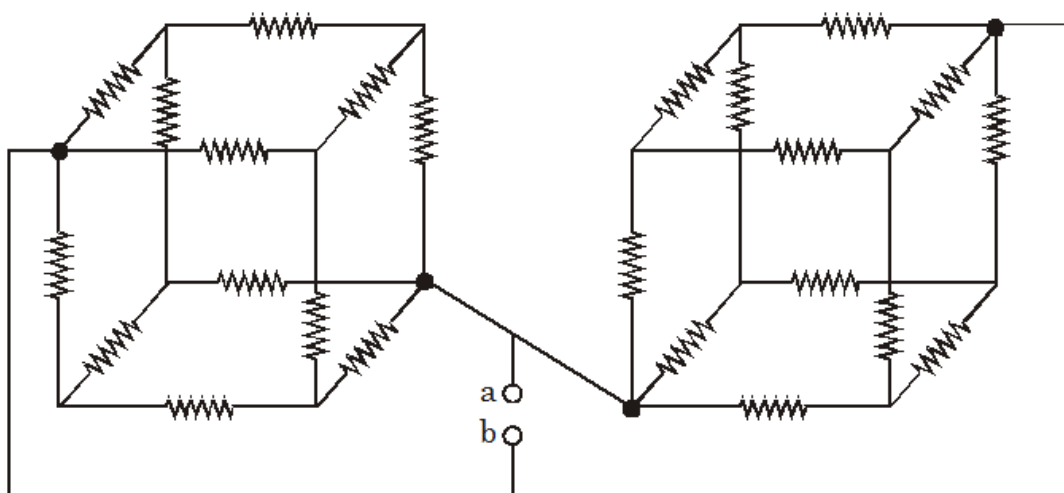
BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

1. Find the equivalent resistance (in Ω) between terminals a and b of the circuit shown in figure. Each resistor has a value of 1Ω .



(A) $\frac{5}{3}\Omega$

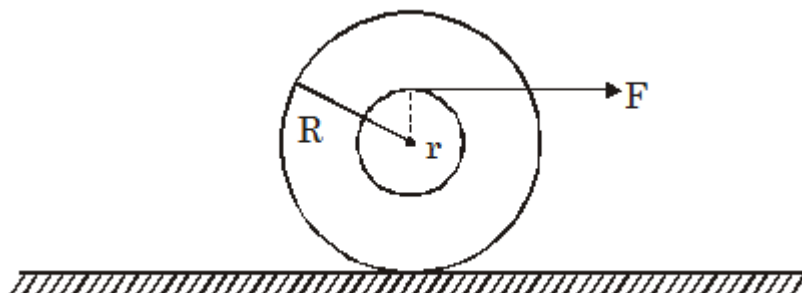
(B) $\frac{5}{6}\Omega$

(C) $\frac{5}{12}\Omega$

(D) $\frac{7}{6}\Omega$

Space for Rough Work

2. A spool of mass 'M' has an inner radius 'r' and outer radius 'R' [Moment of inertia is I about CM = MRr]. Many turns of strings is wound on it and pulled on a horizontal rough surface of coefficient of friction ' μ ' as shown in the figure. The maximum value of applied force F for which it will roll without slipping is (force is applied parallel to ground)



- (A) μMg
 (B) $\frac{\mu RMg}{r}$
 (C) $\frac{\mu Mgr}{R}$
 (D) For any value of F it rolls without slipping
3. The volume charge density of a solid sphere of radius a is directly proportional to the depth below the surface. The depth at which the electric field is maximum is :-
 (A) $\frac{a}{2}$ (B) $\frac{a}{4}$ (C) $\frac{2a}{3}$ (D) $\frac{a}{3}$
4. A solid sphere of mass M of radius R is moving with speed $\sqrt{\frac{GM}{R}} = 100\sqrt{10} \text{ ms}^{-1}$ in free space. Due to application of external impulsive force, it breaks into eight spheres of same radius in such a way that the total mechanical energy before breaking and after breaking of sphere remains same. The speed of each sphere when separation between them is very large (assume at that instant speed of each sphere is same) is
 (A) 100 m/s (B) 50 m/s (C) $50\sqrt{2}$ m/s (D) $100\sqrt{2}$ m/s

Space for Rough Work

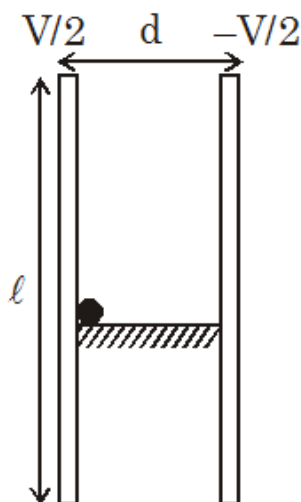
SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. A particle 'P' of mass $\frac{20}{\sqrt{\pi}}$ gm & charge $-q$ was moving in a circular path of radius $\pi^{3/2}$ m with angular velocity $\frac{\pi}{2}$ rad/sec. At centre 'O', a fix charge $+\frac{q}{2}$ is present. At $t = 0$, at point 'A', a neutral particle 'Q' having mass $\frac{60}{\sqrt{\pi}}$ gm initially at rest collide and stick to particle 'P' and simultaneously charge at center O was also neutralized by some mechanism. Then choose the correct statement(s) :-
 (A) Value of q is $\frac{100}{3} \mu\text{C}$
 (B) Just before & after collision between P & Q, linear momentum of the system (P + Q) & angular momentum about O remain conserve
 (C) Just before & after collision, K.E. of P and angular momentum about O decreases.
 (D) Area sweep by OP in between time interval $t = 184$ sec to $t = 216$ sec is 200 m^2 .

Space for Rough Work

2. Two large square metallic plates ($\ell \times \ell$) are separated by a distance d . The potential of plates remain constant at $\frac{V}{2}$ & $-\frac{V}{2}$. The plates are connected by a smooth horizontal insulating platform. A very small metallic ball of mass m and radius r can move on the platform. Assume that when it touches a plate, it instantaneously acquires the same potential as that of the plate. The collision are completely inelastic. $\ell \gg d \gg r$



- (A) The effective charge that flows from left plate to right plate in a complete round trip is $4\pi\epsilon_0 V r$.
- (B) The average current over a large period of time is proportional to $V^{3/2}$.
- (C) The "resistance" to current flow $\left(\frac{V}{i}\right)$ is inversely proportional to V .
- (D) The average power loss is proportional to V^2 .

Space for Rough Work

3. A body of mass m (which is very small than mass of earth M) is to be projected from earth's surface to escape earth's gravitational field. (Radius of earth is R). It is projected in three consecutive steps.

Step-(1) : Minimum sufficient energy or impulse is given to the body so that body moves as satellite of earth at its surface.

Step-(2) : An impulse is given to the body so that it starts to move on an elliptical path in such a manner that its maximum and minimum distances from earth are $4R$ and R respectively.

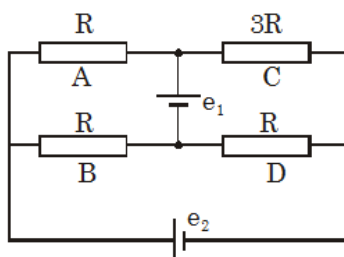
Step-(3) : An impulse is given to the body, when it is at a maximum distance from earth so that it just escapes from earth's gravitational field.

Neglect effect of any other body. Select correct statements.

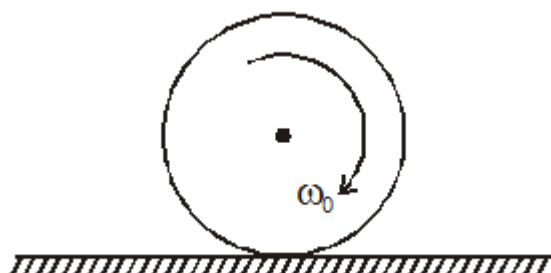
- (A) If body was projected from earth surface to escape from its field, more energy is required than the method described.
- (B) If body was projected from earth surface to escape from its field, same energy is required as in the method described.
- (C) If body was projected from earth surface to escape from its field, less impulse is required than the method described.
- (D) If body was projected from earth surface to escape from its field, same impulse is required as the method described.

Space for Rough Work

4. A circuit is shown in the Figure. EMF of battery e_1 is 1.5 V. EMF of battery e_2 is 3V. Internal resistance of the batteries can be neglected.



- (A) Battery e_1 as well as e_2 are supplying energy to the circuit.
- (B) Battery e_1 is gaining energy but e_2 is supplying energy to the circuit.
- (C) The heat dissipated in resistance C is $\frac{3}{4}$ times the heat dissipated in resistance B
- (D) The current in both the resistances on the left side is the same.
5. A disc is given an initial angular velocity ω_0 and placed on a rough horizontal surface as shown in figure. The quantities which will not depend on the coefficient of friction is/are ($\mu > 0$) :



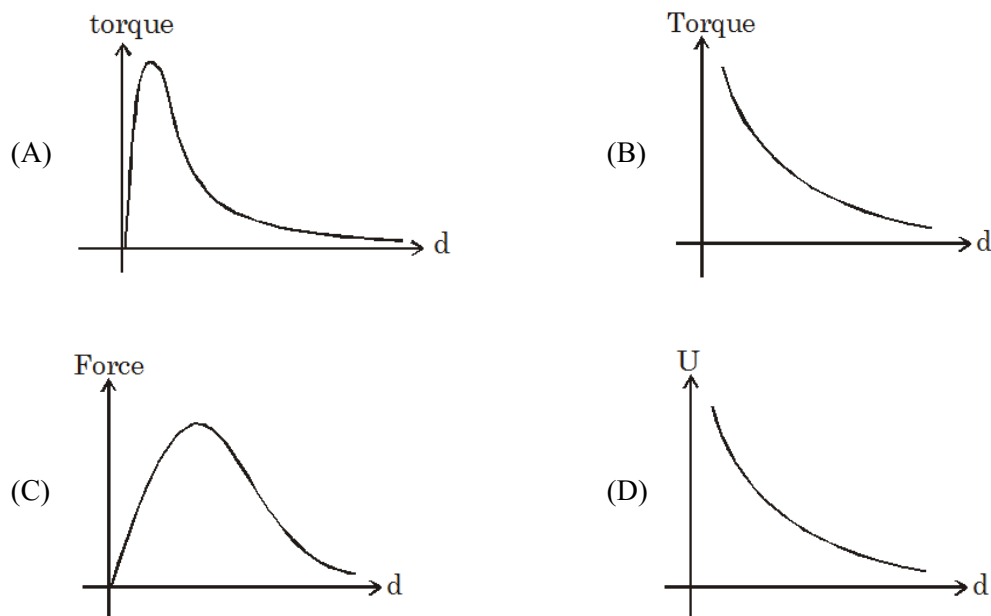
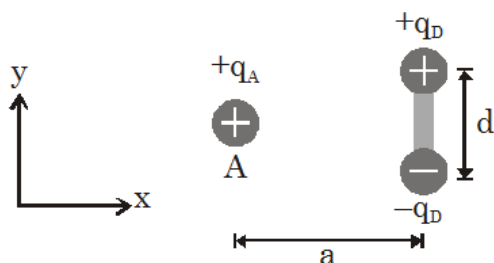
- (A) The time until rolling begins
- (B) The displacement of the disc until rolling begins
- (C) The velocity when rolling begins
- (D) The work done by the force of friction.

Space for Rough Work

6. A very small earthed conducting sphere is at a distance a from a point charge q_1 and at a distance b from a point charge q_2 ($a < b$). At a certain instant, the sphere starts expanding so that its radius grows according to the law $R = vt$, (v is very small). Assuming that the point charges and the centre of the sphere are at rest, and in due time the initial point charges get into the expanding sphere without touching it (through small holes). Then the possible value of current through the connecting of earthing conductor is/are:

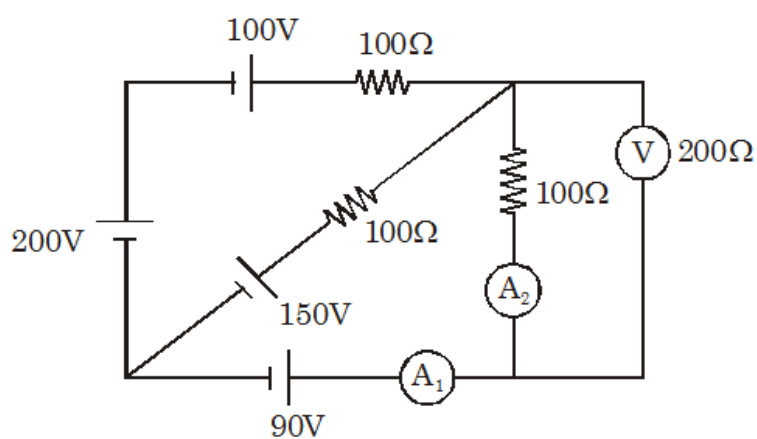
- (A) $I = -v \left(\frac{q_1}{a} + \frac{q_2}{b} \right)$ (B) $I = -v \left(\frac{q_2}{b} \right)$
 (C) 0 (D) $I = -v \left(\frac{q_1}{a} \right)$

7. Figure shows an electric dipole located near a fixed charged particle A that carries charge q_A . The dipole charge is q_0 . Sketch a graph showing how the torque on the dipole (about the centre of dipole), force on the dipole, potential energy of interaction of charge with the dipole changes as the dipole separation d increases symmetrically while the distance a between particle A and the dipole remains constant. d is not necessarily very small.



Space for Rough Work

8. Which of the following options are correct (ammeters are ideal).



- (A) Reading of A_1 is 1.16 A (B) Reading of voltmeter V is 77.14 V
 (C) Reading of ammeter A_2 is 0.77 A (D) Reading of voltmeter is 20 V

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

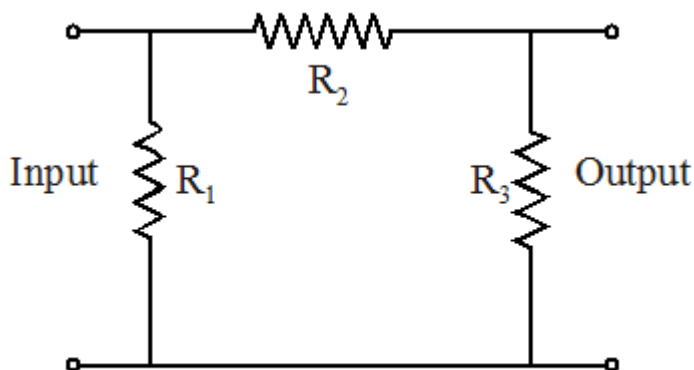
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. Little Prince lives on the spherically symmetric asteroid B-612 (Fig.). The density of asteroids including B-612 is $500\pi \text{ kg/m}^3$. Assume that the asteroid does not rotate. The Little Prince noticed that he felt lighter whenever he walked quickly around his asteroid. In fact, he found that he became weightless and started to orbit the asteroid like a satellite whenever he speeded up to 2m/s. (Take $G = \frac{20}{3} \times 10^{-11}$ SI units. $\pi^2 \approx 10$). Radius of asteroid (in km) is.



Space for Rough Work

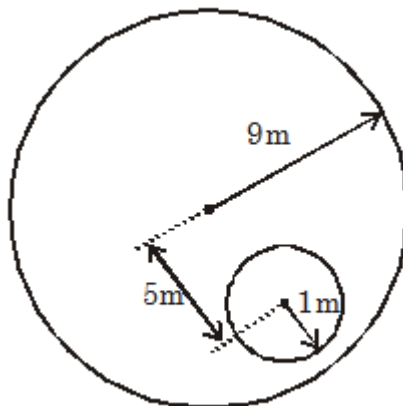
2. If at the input circuit (Figure) a voltage $U_1 = 100$ V, is applied the potential difference at the output $U_2 = 40$ V. If at the output circuit a voltage $U_3 = 60$ V is applied, the potential difference at the input will be $U_4 = 15$ V. resistor $R_2 = 60$ ohms. Find electrical resistance of resistors R_1 and R_3 (in Ohm) and fill $R_1 + R_3$ in OMR sheet.



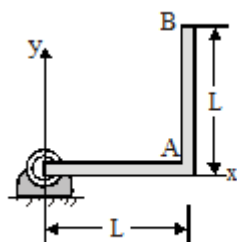
3. Consider two spheres of radius a , immersed deep in the sea and held quite far apart. If the potential difference between them is V , and current measured to be I , then conductivity of sea water is given by σ S.I. unit. Find the value of σ . [Given : $I = 3$ A, $V = 10$ Volt & $a = 1$ cm]

Space for Rough Work

4. A point charge 2mC is taken from infinite to centre of spherical cavity of radius 1m inside metallic solid sphere of radius 9m as shown in figure. Then work done by electric forces on point charge is $\alpha(2\beta \times 10^7)\text{J}$. Here α is for sign, if work is +ve then take value of α is zero otherwise take its value '2' then find value of $(\alpha + \beta + \gamma)$? (β & γ are integers in between 0 to 9)



5. Consider an "L-shaped rod" of uniform mass density, hinged at point 'O' is held at rest initially. The mass of the rod is ' $m = 1\text{kg}$ '. The earth's gravitational field is assumed to be uniform and directed downwards. Assume no friction at hinges. The rod is now released. Find the maximum magnitude of angular speed of the rod (in rad/s). (Take : $L = 1\text{m}$, $\sqrt{10} = 19/6$)

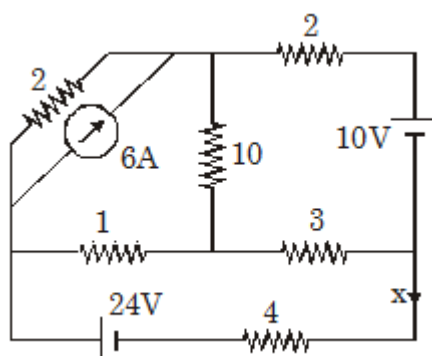


Space for Rough Work

6. Determine the current x (in A) in the 4Ω resistance of the circuit shown in figure. The



represents constant current carrying source of 6A.



Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Match the list : (Atomic number of Ni = 28, Ir = 77, Pt = 78, Co = 27)

List- I

(Co-ordination compounds)

(P) $K_2[NiBr_2(NO_3)_2(en)]$

(Q) $[IrCl(SCN)(H_2O)(NH_3)_3]$

(R) $[Pt(gly)_2(NO_2)_2]$

(S) $[Co(gly)(H_2O)_2(NH_3)_2][Cu(NO_2)_4]$

List- II

(Characteristics of coordination compounds)

(1) It exhibit structural isomerism

(2) High spin complex

(3) Low spin complex

(4) Electron migrates from (n-1)d subshell of central metal cation to its higher energy subshell

(5) Total number of optically active isomers are equal to two.

The CORRECT option is -

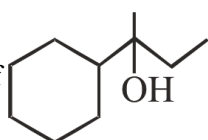
(A) $P \rightarrow 5$; $Q \rightarrow 1,3,4,5$; $R \rightarrow 1,3$; $S \rightarrow 1,3,4$

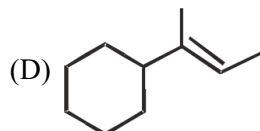
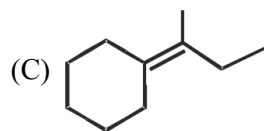
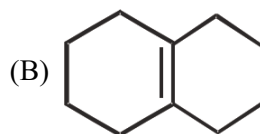
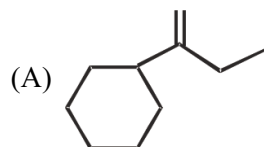
(B) $P \rightarrow 2,5$; $Q \rightarrow 3,4,5$; $R \rightarrow 3,5$; $S \rightarrow 1,2,5$

(C) $P \rightarrow 2,4$; $Q \rightarrow 3,4,5$; $R \rightarrow 1,3,4$; $S \rightarrow 1,4$

(D) $P \rightarrow 1,2$; $Q \rightarrow 3,5$; $R \rightarrow 1,2$; $S \rightarrow 1,4,5$

Space for Rough Work

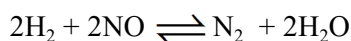
2. Which of the following is not the product of acidic dehydration of  ?



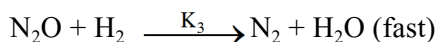
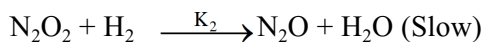
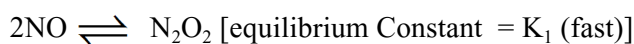
3. For the nuclear reaction ; ${}^A_Z X \longrightarrow {}^{A-8}_{Z-4} Y$; $t_{1/2} = 1600$ yrs.
If initial activity was 10^7 dps, how many α -particles will be emitted per second after 4800 years ?

- (A) $1.25 \times 10^6 \text{ s}^{-1}$ (B) $2.5 \times 10^6 \text{ s}^{-1}$ (C) $1.25 \times 10^7 \text{ s}^{-1}$ (D) $5 \times 10^7 \text{ s}^{-1}$

4. For the reaction:



the following mechanism has been suggested :



According to following mechanism is correctly matched:

(A) $-\frac{1}{2} \frac{d[\text{H}_2]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = K_2 \times K_3$

(B) $-\frac{1}{2} \frac{d[\text{NO}]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = K_2 \times K_1$

(C) $\frac{d[\text{N}_2]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = K_2 \times K_1$

(D) $-\frac{1}{2} \frac{d[\text{NO}]}{dt} = K[\text{NO}]^2 [\text{H}_2]$ Where $K = K_2 \times K_1$

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

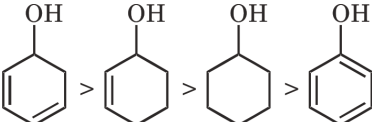
1. Which order is correct :

(A) Nucleophilicity (In DMSO) : $\text{F}^\ominus > \text{Cl}^\ominus > \text{Br}^\ominus > \text{I}^\ominus$

(B) Leaving group tendency : $\text{N}_2 > \text{O}^\ominus - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3 > \text{OH}^\ominus$

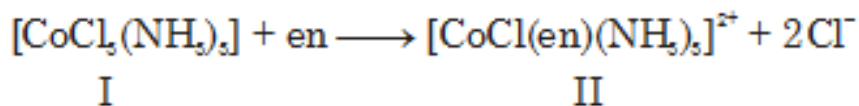
Rate of reaction (Electrophillic addition reaction) :

(C) $\text{OCH}_3 > \text{CH}_3 > \text{NO}_2$

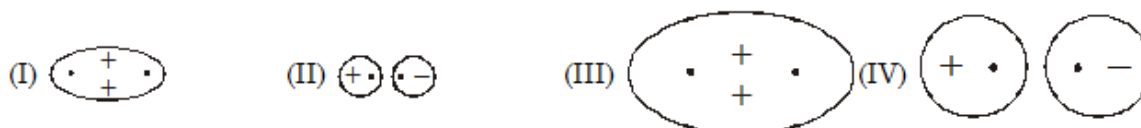
(D) Rate of reaction (Dehydration) : 

Space for Rough Work

2. The correct statement(s) for the following reaction is/are.



- (A) If complex compound-I is a fac isomer, then complex ion-II will be fac-isomer w.r.t. NH_3 .
 (B) If the complex compound-I is mer isomer, then two isomers are produced.
 (C) If complex compound-I is a fac isomer, then complex ion-II exhibit optical activity.
 (D) If complex compound-I is a mer isomer then complex ion-II does not exhibit optical activity
3. Consider LCAO of 1s and 2s orbitals of Li atoms to form Li_2 molecule. Which produce given four molecular orbitals :-

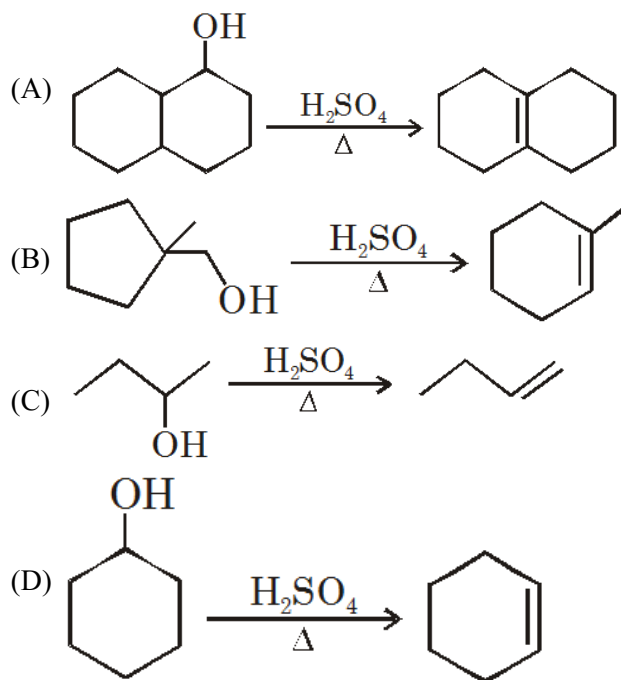


Which of the following statement is/are correct?

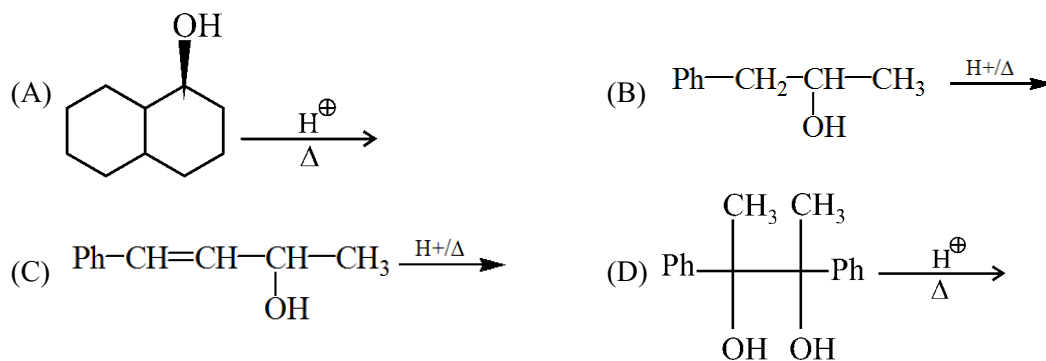
- (A) M.O. IV is vacant orbital in Li_2
 (B) Electron pair in M.O. III is responsible for existence of Li_2
 (C) M.O. II and M.O. IV have common nodal plane
 (D) Addition and removal of electron take place from M.O. III when Li_2 undergoes in either Li_2^+ or Li_2^- formation

Space for Rough Work

4. The reaction(s) is/are correctly matched with their major product :



5. In which of the following reaction(s) rearrangement of carbocation takes place.



Space for Rough Work

6. A sample of ${}_{92}^{238}\text{U}$ when kept at 298 K in a closed vessel shows α -decay and β -decay to give ${}_{82}^{206}\text{Pb}$. After t hour the volume of the gas collected in the vessel at 1 atm, 273 K is found to be 89.6 ml. Then which of the following is/are correct for the nuclear process and the products of the process.
- (A) Mass of Pb obtained in time t hour = 103 mg.
- (B) If the temperature is increased by 10°C then the time take for the same amount of gas to be produced will be $\frac{t}{2}$ hour.
- (C) α particles show more ionisation power than β particles.
- (D) γ radiations undergoes no deviation while traveling through electric or magnetic field.
7. Potassium crystallizes in a body centred cubic lattice with edge length $a = 5.0\text{\AA}$. Then choose the correct statements in given option -
- (A) The number of nearest neighbour to any K atom is 8
- (B) The number of next nearest neighbour to any K atom is 6.
- (C) Distance between nearest neighbours is $4.33\text{\AA}$
- (D) Distance between next nearest neighbours is $5.0\text{\AA}$
8. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult law. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2 . Pick out the incorrect statements among the following.
- (A) Boiling temperature of mixture is always less than boiling temperature of acetone
- (B) Boiling temperature of Azeotropic mixture is always less than boiling temperature of pure CS_2
- (C) When a small amount of CS_2 (less volatile component) is added to excess of acetone boiling point of resulting mixture increases
- (D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

- As per C.F.T. Cl-Mn-Cl angle in $[\text{MnCl}_4]^{2-}$ is $109^\circ 28'$ what is the angle between e-orbital of central metal and the ligand is –
 (If your answer is $84^\circ 32'$ then you have to fill 84.32 in OMR)
- Consider given reaction (py = pyridne) :

$$3[\text{Co}_2(\text{CO})_8] + 12 \text{ py} \rightarrow 2[\text{Co}(\text{py})_6] [\text{Co}(\text{CO})_4]_2 + 8\text{CO}$$
 - Total number of valence shell electrons on both central metals in $\text{Co}_2(\text{CO})_8 = \text{P}$
 - Number of moles of metal-carbon linkage in each mole of product complex = Q
 - Difference between oxidation number of cobalt in cationic & anionic part of product complex = R.
 Find sum of P, Q, R
- Number of ores which produce CO_2 on thermal decomposition -
 dolomite, lime stone, calamine, bauxite, haematite, siderite, chalcocite, galena, malachite

Space for Rough Work

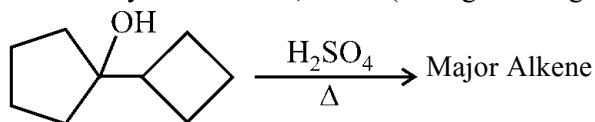
4. Consider two reactions at 27°C

(I). $A \rightarrow B$; (I order kinetics)

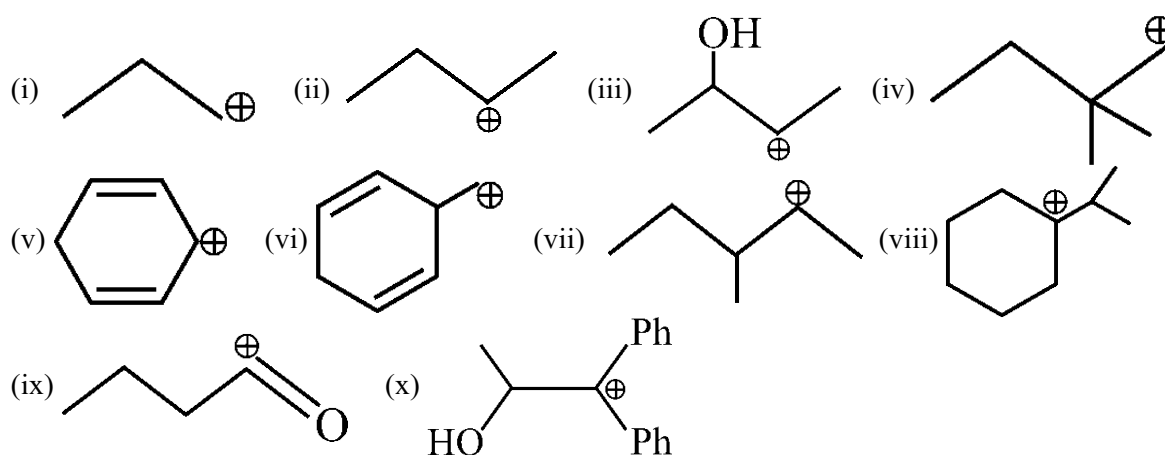
(II). $X \rightarrow Y$; (Zero order kinetics)

If both A & X each having initial concentration 0.1M & same half life period. Then find ratio of rate constants for I & II reactions. [$\ln 2 = 0.7$]

5. How many number of 1,2-shift (during rearrangement of carbocation) for the following reaction :



6. How many carbocation can undergo rearrangement :



Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $y = e^{a \sin^{-1} x}$, then $(1 - x^2)y_{n+2} - (n^2 + a^2)y_n$, $n \in \mathbb{N}$ is equal to (where 'a' is constant and y_n represents $\frac{d^n y}{dx^n}$) -
 (A) $(n^2 + n + 1)xy_{n+1}$ (B) $(2n + 1)xy_{n+1}$
 (C) $(n^2 - n + 3)xy_{n+1}$ (D) none of these
2. If number of points of discontinuity of $f(x) = \text{sgn}(\cos 5x)$ is equal to the number of points of non-derivability of $g(x) = \{n + m \sin x\}$ where $x \in (0, \pi)$, $n, m \in \mathbb{I}$ then the value of m is
 (Note : $\{x\}$ denotes fractional part function of x and $\text{sgn}(x)$ denotes signum function of x)
 (A) 1 (B) 2 (C) 3 (D) 4
3. Let $f(x)$ is a differentiable function everywhere and suppose $\lim_{x \rightarrow \infty} (f'(x) + 3f(x)) = 12$, then $\lim_{x \rightarrow \infty} f(x)$ is equal to
 (A) 2 (B) 4 (C) 6 (D) 12
4. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2$
 (A) 0 (B) 1 (C) 2 (D) 3

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

- The angle between the pair of curves $x^2 - y^2 = a^2$, $x^2 + y^2 = \sqrt{2}a^2$ is
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
- The value of $\left[\sqrt{\lambda^{2017}} + \sqrt{\lambda^{2017} + 1} \right] \forall \lambda \geq 2, \lambda \in \mathbb{N}$ (where $[.]$ is greatest integer function) is equal to-
 (A) $\left[\sqrt{4\lambda^{2017} + 2} \right]$ (B) $\left[\sqrt{4\lambda^{2017} + 1} \right]$
 (C) $\left[\sqrt{4\lambda^{2017} + 3} \right]$ (D) $\left[\frac{1}{\sqrt{\lambda^{2017} + 1} - \sqrt{\lambda^{2017}}} \right]$

Space for Rough Work

3. Let $f(x)$ be a polynomial function satisfying the following conditions
 $\lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0$, $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = -1$ and $f(0) = 0$ (Note : $[k]$ denotes greatest integer function less than or equal to k)
- (A) The number of points of discontinuity of $g(x) = [f(x)]$ in $[0,3]$ is 5
 (B) The number of points of non-derivability of $h(x) = |f(|x|)|$ is 3
 (C) The number of points of discontinuity of $g(x) = [f(x)]$ in $[0,3]$ is 2
 (D) The number of points of non-derivability of $h(x) = |f(|x|)|$ is 1
4. Let $(1 + \sqrt{5})^n = \alpha_n + \beta_n \sqrt{5}$ where α_n and β_n are integers, then choose the correct option(s).
- (A) $\lim_{\theta \rightarrow 0} \frac{\tan^{-1} \left(\frac{\alpha_{n+1} - \beta_{n+1}}{3} \right) \theta}{\sin^{-1} (\beta_n \theta)} = \frac{4}{3}$ (B) $\lim_{\theta \rightarrow 0} \frac{\tan^{-1} \left(\frac{\alpha_{n+1} - \beta_{n+1}}{3} \right) \theta}{\sin^{-1} (\beta_n \theta)} = \frac{3}{4}$
 (C) $\left(\frac{\alpha_n}{2^n} \right)^2 - 5 \left(\frac{\beta_n}{2^n} \right)^2 = (-1)^n$ (D) $\left(\frac{\alpha_n}{2^n} \right)^2 - 5 \left(\frac{\beta_n}{2^n} \right)^2 = (-1)^{\frac{n}{2}}$
5. The length of the segment of the tangent to the curve $x = \frac{a}{2} \log \left(\frac{a + \sqrt{a^2 - y^2}}{a - \sqrt{a^2 - y^2}} \right) - \sqrt{a^2 - y^2}$ contained between the y-axis and point of tangency is
- (A) $2a$ (B) $\frac{a}{2}$ (C) a (D) $3a$

Space for Rough Work

6. Let f be a polynomial function satisfying $f(x^2) - xf(x) = x^4(x^2 - 1) \forall x \in \mathbb{R}$. Then, which of the following is correct
- (A) f is an even function
- (B) f is an odd function
- (C) $\lim_{x \rightarrow \infty} \frac{f(x)}{x^3} = 1$
- (D) $\lim_{x \rightarrow \infty} \left(\frac{f(x)}{x^2} - x \right)$ exist and is equal to a non-zero quantity
7. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2([x] - \{x\})$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = 2 \left[\frac{x}{2} \right] - \frac{x}{2}$, then which of the following holds good ?
(where $[.]$ and $\{.\}$ denotes greatest integral function & fraction part function respectively)
- (A) Area of triangle formed by $y = f \circ g(x)$, $x = 0$, $y = 0$ is 2.
- (B) Area of triangle formed by $y = g \circ f(x)$, $x = 0$, $y = 0$ is 2.
- (C) $f(x)$ is bijective function.
- (D) $g(x)$ is bijective function.
8. Let $f(x) = \begin{cases} x^2 - 2x + 3 & ; x \in (-2, 0) \\ x^3 - x^2 + 2 & ; x \in [0, 2] \\ 3 - 2x & ; x \in (2, 6] \end{cases}$, then
- (A) $f(f(x))$ is discontinuous at 4 points.
- (B) $f(f(x))$ is discontinuous at 3 points.
- (C) $f(f(0)) = -1$
- (D) $f(f(2)) = -9$

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

- If $f(2016) = \frac{\pi}{2}$, $f(x) = f(2x) \forall x \in \mathbb{R}$ and $f(x)$ is a continuous function for all $x \in \mathbb{R}$, and $a = \lim_{x \rightarrow 0} \frac{\cos f(x) - \sqrt[3]{\cos f(x)}}{\sin^2 x}$ and b is number of points of $g(x) = \max\{\tan x, \cot x\}$ in $x \in (-\pi, \pi)$ where it is non-differentiable, then $\left(\frac{a+b}{4}\right)$ is equal to
- The value of $\left| \lim_{x \rightarrow 0} \frac{1 - (\cos 2x)^3 (\cos 5x)^5 (\cos 7x)^7 (\sec 4x)^9 (\sec 6x)^{11}}{4x^2} \right|$ is-
- If $\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1}{\sqrt{1+x}} - \frac{1-ax}{1-bx} \right)$ exists and has the value equal to ℓ then value of $\frac{1}{4} \left(\frac{1}{a} - \frac{3}{b} - 32\ell \right)$ is equal to :

Space for Rough Work

4. If $y = mx + c$ is bitangent (a line is tangent to the curve at two distinct points) on the curve $y = x^4 + 2x^3 - 11x^2 - 13x + 35$, then the value of $\frac{|m+c|}{5}$ is
5. If n is even natural number, then the number of real roots of the equation $1 + 2x + 3x^2 + \dots + (n+1)x^n = 0$ is
6. The value of $\lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{4\sqrt{2} - (\cos x + \sin x)^5 - 10 \cos x + 10 \cos x \sin 2x}{1 - \sin 2x} \right)$ is
-

Space for Rough Work

Space for Rough Work